

Troubleshooting Guide

Queensland Reds | Computer Vision Software

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Frequently Asked Questions (FAQ)

Note: The project was run on a development platform called Visual Studio Code (VS Code). While you can run the program without VS Code, if you'd like to see the code for the programs, this is the integrated development platform (IDE) we recommend and is what is shown in our examples.

Q: Nothing happens when I run the python file to start the program

A: Are you running the correct file - main.py?

In Finder or File Explorer, double click the main.py file to run it.

If you're in VS Code:

On the left-hand column, double click `main.py` and then click the play button on the top right-hand side to run the file. When you click play/run it should say `<filepath>/src/main.py` in the terminal

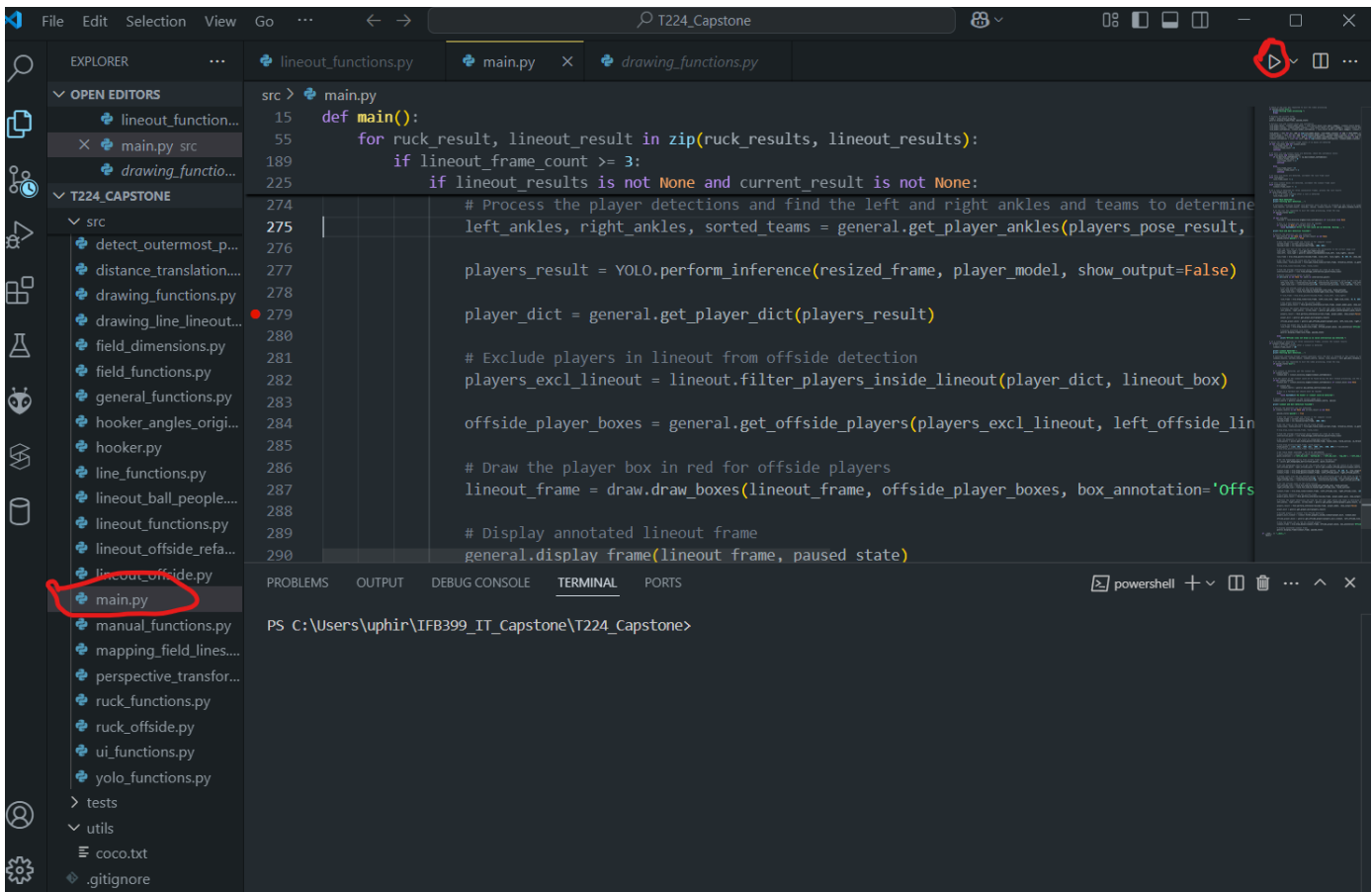


Figure 1.1: I

A: Check your Desktop

Sometimes when you run the file, if you have any other windows or applications open on your computer, the popup where you select your video appears underneath everything on your desktop. To quickly navigate to your desktop on Windows, click the very far right part of your navigation bar (in this instance it is in the bottom right corner of the screen)

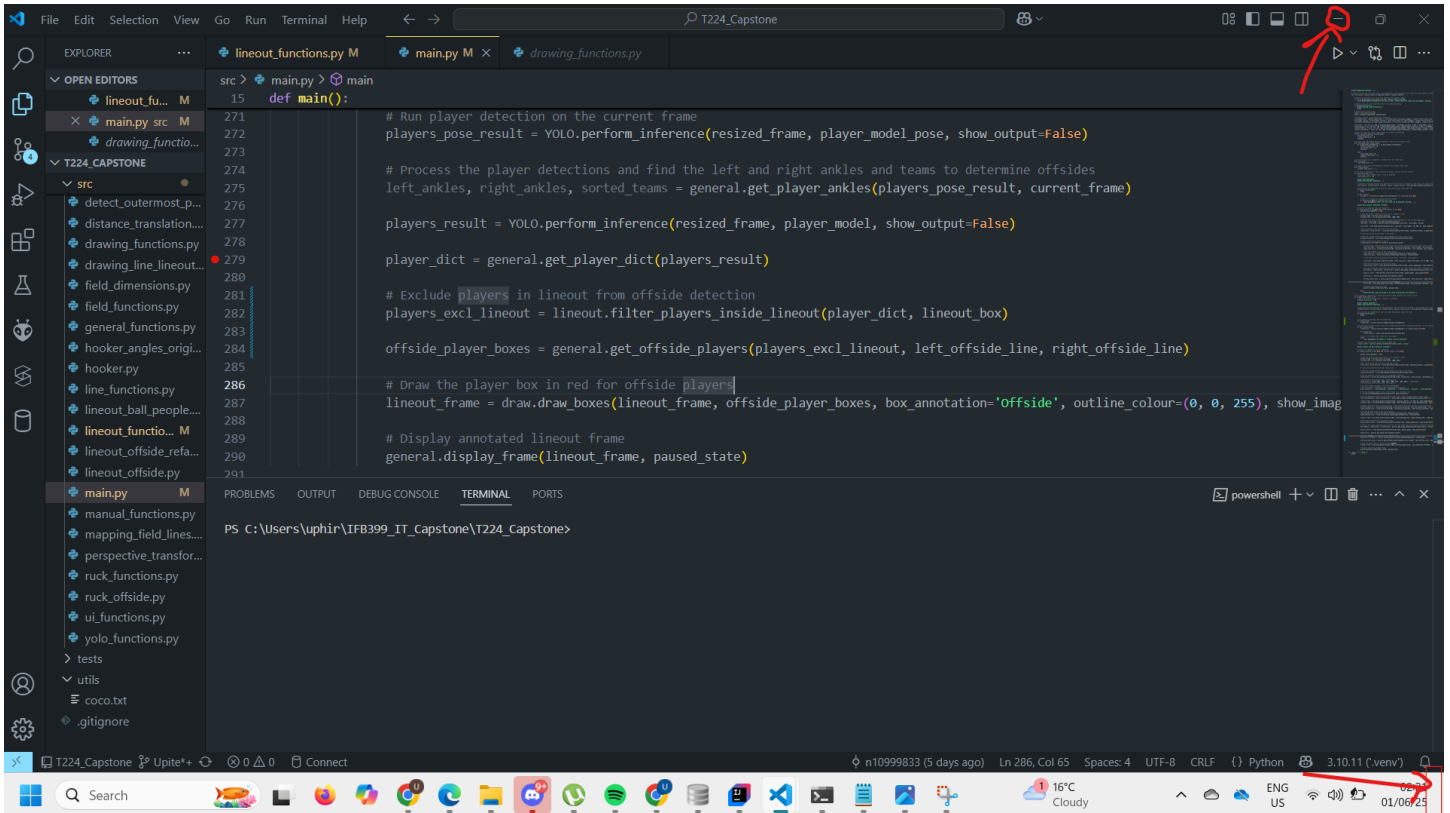


Figure 1.2: In Windows, the far corner of your navigation bar will take you to your desktop. Otherwise just minimize your pages using the '-' in the top-right until you can see the desktop.

You should then be able to see a window with the heading 'Select File'.

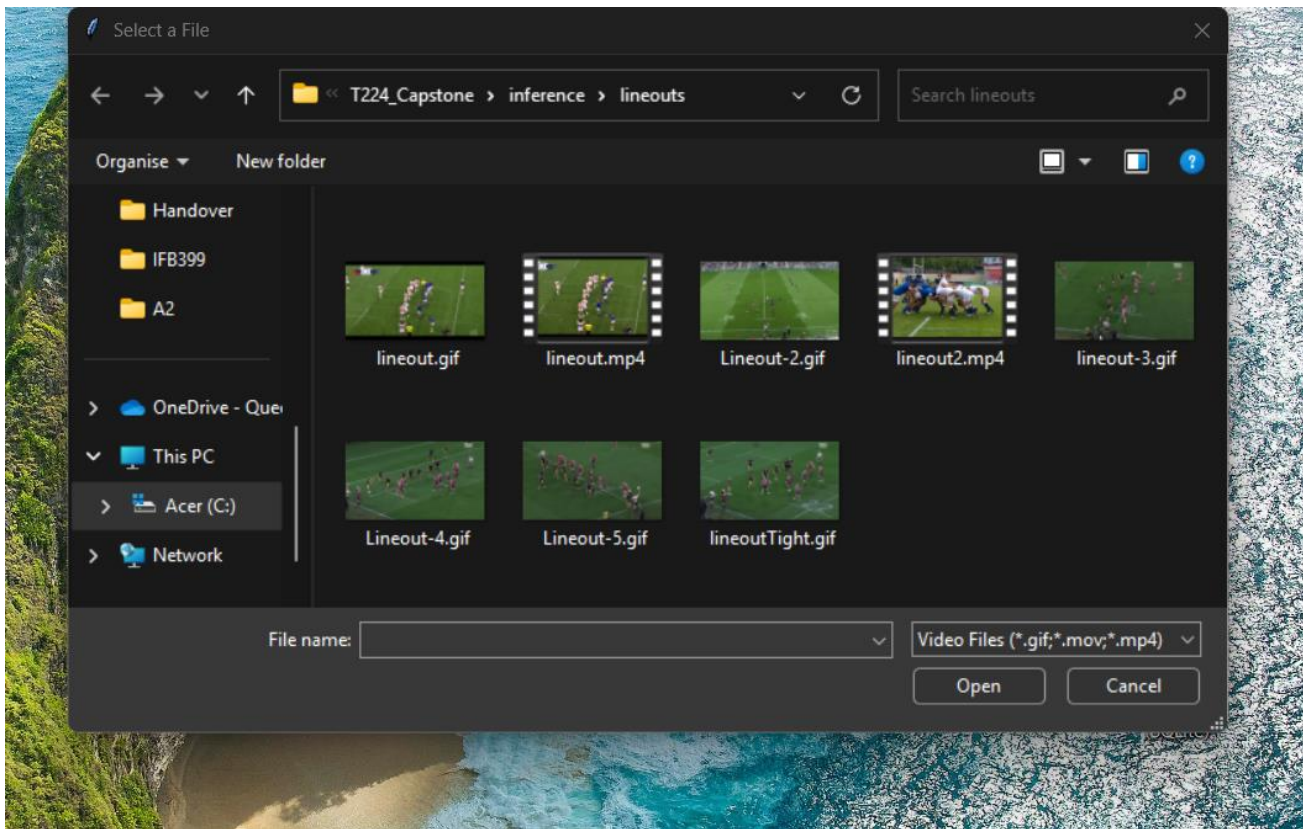


Figure 2.3: Select file window pop-up allows you to select footage to test

Q: Program appears to run then crashes

A: Open in an IDE

If the program launches, then crashes, you might get an error message in the terminal of the development platform or IDE you are in. If you tried to launch the program (through File Explorer or Finder), the error message may be abstracted from view. We recommend opening in an IDE, namely VS Code, to see what may have caused the error.

A: Check installations

[Click here to jump to section.](#)

Q: The field lines aren't drawing or aren't drawing precisely

A: Your threshold selection might be too high or too low

When you run main.py and select a file, you will have to set the threshold so that the field lines can be accurately seen. Setting your threshold too low or too high may impact the computer vision visualization. By default, threshold is set at 165, but that may not be the best setting depending on the video you've selected. Change the threshold by moving the slider at the top. Then, select any key to confirm your choice and begin video processing.

Ideally, you want the threshold set so you can clearly see the line marking on the field (see Figure 2.1).



Figure 2.1: Threshold set to an ideal amount where field markings are clearly visible.

What you want to avoid is setting your threshold too low, which will not provide enough detail for video processing (see Figure 2.2).

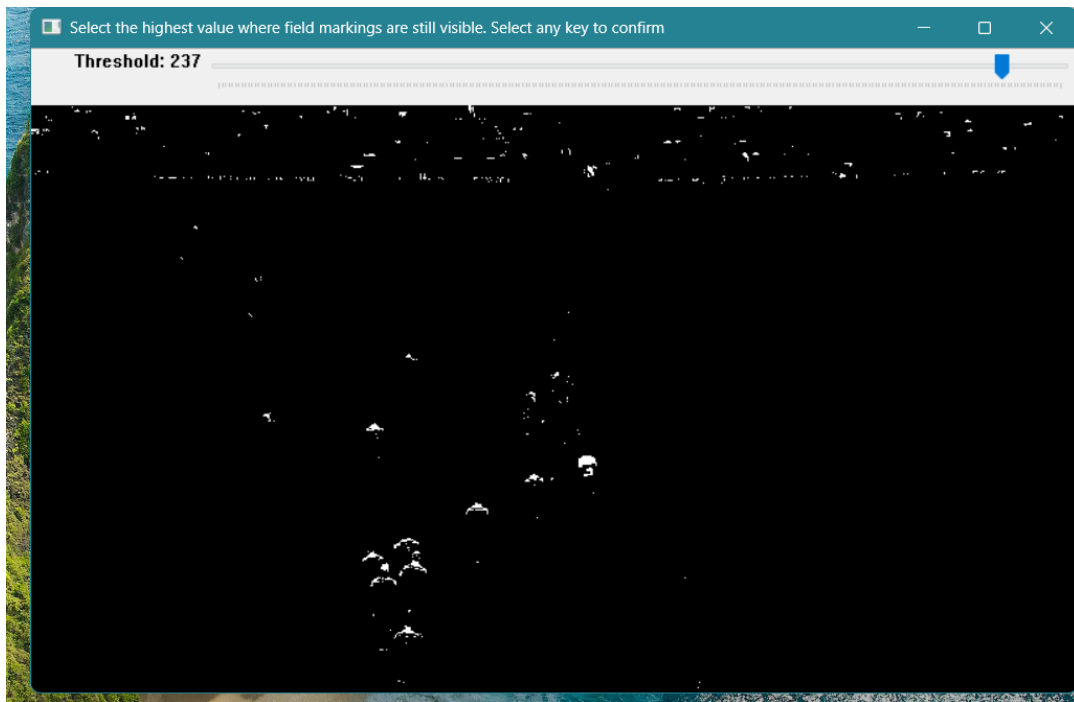


Figure 2.2: Threshold is set too low.

Nor should you set it too high, as this may result in the system detecting more obscure details or anomalies (see Figure 2.3).



Figure 2.3: Threshold is set too high.

Q: I'm closing the video window, but it keeps reappearing

A: Hit 'q' to exit

If the analysis is finished and you'd like to close the analysis window that pops up but find that it reappears every time, instead of hitting the 'x' on the window or closing VS Code, hit the 'q' on your keyboard to stop it. In the terminal, you will get a message saying, 'Exit video processing'.

Q: When I try to run the program, it says I have items or modules missing

If you attempt to run the program within VS Code, and you get this error, here are some things you can check.

A: Run the installation file for your system

In the Installation folder, there should be an install file:

1. A bash script (.bh) for a Windows installation
2. A shell script (.sh) for Mac and Linux installation

Before installing, please read the file providing information about installation instructions (.txt) for your computer's operating system. When you're done, double click the script file to run, or right-click and select "Open" or "Run".

When the script has finished, it will notify you in the command line of success or failure. If successful, the modules/packages should be installed. To check please refer [to the next section](#).

If you are a Mac or Linux user, the installation file will also install Homebrew, package manager, that will download python.

A: Confirm that your computer has downloaded modules from installation

If you need to manually install any packages, see this section for command line prompts.

After running the installation file for your computer, you should have the following:

- python3: Python interpreter
- python3-pip: Python package installer
- python3-tk: Python Graphical User Interface (GUI) kit
- opencv-python: OpenCV bindings for image and video processing (core modules only)

- `opencv-contrib-python`: OpenCV with **extra modules**, including experimental and advanced features
- `ultralytics`: Official YOLOv8 library for object detection, segmentation, tracking, and more
- `matplotlib`: Library for creating static, interactive, and animated plots and charts
- `pydantic`: Data validation and settings management using Python type annotations
- `scikit-learn`: Popular machine learning library with tools for classification, regression, clustering, etc.

Check installed packages in computer or IDE terminal

To display the list of your installed modules, type:

```
pip list
```

Important: the installation should have downloaded pip and Python for you, if you are missing pip or python or are unsure if those modules were installed, please read the following section.

Check that Python and its package manager 'pip' are installed

```
python -version
```

If installed, you will see: `python <version>`

If not installed, you will see a 'not found' error

```
pip -version
```

Update pip or Python

To update pip:

```
python -m pip install --upgrade pip
```

To update Python (macOS):

```
brew update
```

```
brew upgrade python
```

To update Python (Windows): Officially, you'll have to go to the organisations website and download a new version, or you can try use the interpreter in VS Code

A: Check the version compatibility of the installed packages

The computer vision system currently runs on YOLO ultralytics version 8.3.102. All other modules will need to be compatible with this version of Ultralytics.

Note: if you have upgraded a package where the version is a dependency, then you may have installed an earlier version. In this guide, packages and modules are used interchangeably.

The following compatibilities are required:

- opencv-python, version 4.6.0 or earlier
- numpy, any version between 1.23.0 and 2.1.1
- Most versions of opencv-contrib-python are fine; recommended 4.8.0.74
- python, version 3.8 or later; recommended is 3.10 or 3.11

How to check versions in terminal

For instructions on how to access the terminal/command line in VS Code or on your computer: [click here](#).

To check a specific module version, type:

```
pip show <name of module>  
e.g.: pip show ultralytics
```

How to install a package

If you need to install a specific version of a package, type:

```
pip install <name of module>  
e.g.: pip install ultralytics
```

How to install an alternate version

If you need to install a specific version of a package, type:

```
pip install <name of module> == <version>
```

```
e.g.: pip install opencv-contrib-python==4.8.0.74
```

```
pip install numpy==1.24.4
```

How to uninstall a package

To prevent any conflicts, you may choose to uninstall a version before reinstalling a different version. To do this, you have to uninstall the module.

To uninstall, type:

```
pip uninstall <name of module>
```

```
e.g.: pip uninstall ultralytics
```

How to update package

Otherwise, if you'd just like to update your package, type:

```
pip install --upgrade <package-name>
```

Q: How do I run code commands in terminal?

The project was run on a development platform called Visual Studio Code (VS Code). Some of the examples where terminal is shown, it is in VS Code however, most of the commands can be entered into Windows, Linux or Mac command line terminal.

A: Open...

Windows PowerShell/Command Prompt

Method 1: Using the Start Menu

1. Click the **Start** button (Windows icon) in the bottom-left corner of your screen.
2. Type **cmd** or **Command Prompt** in the search bar.

3. Click on the **Command Prompt** app that appears in the search results.

Method 2: Using the Run Dialog

1. Press **Windows key + R** on your keyboard to open the Run dialog box.
2. Type **cmd** into the box.
3. Press **Enter** or click **OK**.

Method 3: From File Explorer

1. Open **File Explorer** (folder icon on the taskbar or press **Windows key + E**).
2. Click on the address bar at the top.
3. Type **cmd** and press **Enter**.
 - a. This will open Command Prompt in the current folder location.

Open as Administrator (if needed)

1. Search for **Command Prompt** in the Start Menu.
2. Right-click the **Command Prompt** app.
3. Select **Run as administrator**.
 - a. This gives Command Prompt elevated permissions needed for some commands.

Mac/Linux Terminal

Method 1: Using Spotlight Search

1. Press **Command (⌘) + Spacebar** to open Spotlight Search.
2. Type **Terminal**.
3. Press **Enter** when Terminal appears as the top result.

Method 2: Using Finder

1. Open **Finder** (the smiling face icon in the Dock).
2. Go to **Applications** in the sidebar.
3. Open the **Utilities** folder.
4. Double-click **Terminal** to open it.

Method 3: Using Launchpad

1. Click the **Launchpad** icon in the Dock (rocket icon).
2. Type **Terminal** in the search bar at the top.
3. Click the **Terminal** app icon to open it.

Run as Administrator (if needed)

There is no admin mode, per se. Just prefix the command with `sudo` and enter your password when prompted if you need elevated/admin permissions.